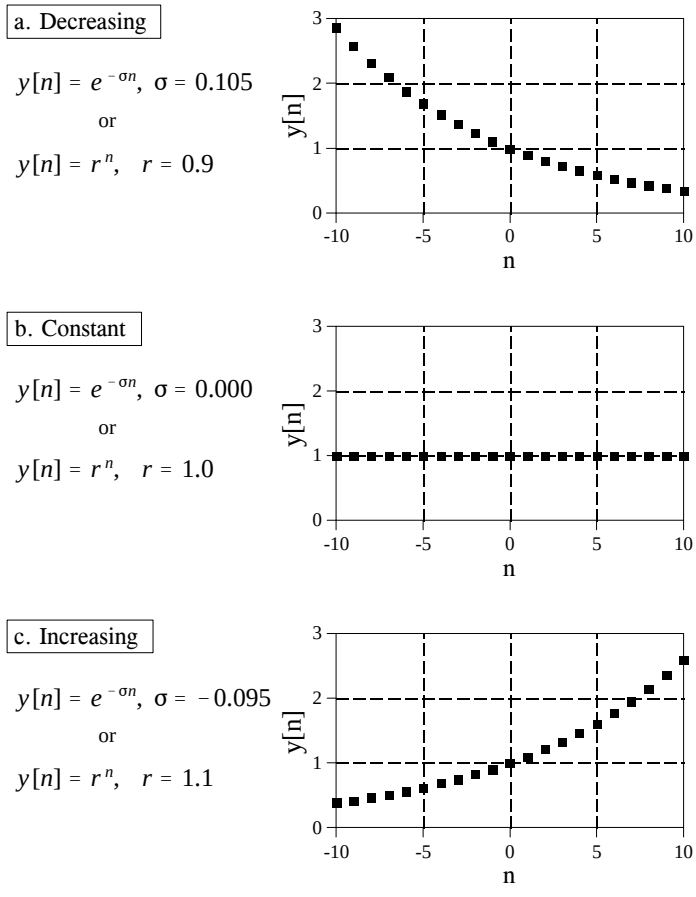

FIGURE 33-1
Exponential signals. Exponentials can be represented in two different mathematical forms. The Laplace transform uses one way, while the z-transform uses the other.

The second expression uses the parameter, r, to control the decay of the waveform. The waveform will decrease if $r < 1$, and increase if $r > 1$. The signal will have a constant value when $r = 1$. These two equations are just different ways of expressing the same thing. One method can be swapped for the other by using the relation:

$$r^n = [e^{\ln(r)}]^n = e^{n\ln(r)} = e^{-\sigma n}$$

where: $\sigma = -\ln(r)$

The second step of converting the Laplace transform into the z-transform is completed by using the *other* exponential form:

$$X(r,\omega) = \sum_{n=-\infty}^{\infty} x[n]\, r^n\, e^{-j\omega n}$$

While this is a perfectly correct expression of the z-transform, it is not in the most compact form for complex notation. This problem was overcome